

Future Paradigms for Sustainable Making

Zeyu Yan
zeyuy@umd.edu
University of Maryland, College Park
College Park, MD, USA

Tingyu Cheng
tcheng32@gatech.edu
Georgia Institute of Technology
Atlanta, GA, USA

Jasmine Lu
jasminelu@uchicago.edu
University of Chicago
Chicago, IL, USA

Pedro Lopes
pedrolopes@uchicago.edu
University of Chicago
Chicago, IL, USA

Huaishu Peng
huaishu@umd.edu
University of Maryland, College Park
College Park, MD, USA

ABSTRACT

This workshop provides the first opportunity for the UIST community to discuss sustainability challenges and opportunities in rapid prototyping. We will discuss key issues such as waste generation from intermediate prototypes, strategies for sustainable materials, circular prototyping (e.g., promoting re-use of components), knowledge sharing infrastructures (e.g., open-sourcing hardware), and so forth. The goal is to identify potential HCI research directions that can foster a more sustainable "making" environment inside of labs and beyond.

CCS CONCEPTS

• Human-centered computing → User interface toolkits.

KEYWORDS

Rapid Prototyping, Sustainability, Reuse, Recycle, Making, Maker Community

ACM Reference Format:

Zeyu Yan, Tingyu Cheng, Jasmine Lu, Pedro Lopes, and Huaishu Peng. 2023. Future Paradigms for Sustainable Making. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3617433>

1 BACKGROUND AND GOAL

Over the past decades, democratized fabrication and electronic toolkits have proven to be a monumental driving force for innovation. Digital fabrication tools (e.g., 3D printers, laser cutters) and electronics toolkits (e.g., Arduino, CircuitPython), have become widespread in research labs and makerspaces. Enabling accessible rapid prototyping has significantly shortened the innovation cycle. However, the ease with which rapid design iterations can be carried out ad hoc and in situ has also led to considerable sustainability challenges that are often overlooked.

The development of new products through rapid prototyping inevitably produces a large number of intermediate prototypes,

such as 3D printed structures and custom PCBs. These intermediate prototypes are created to verify various aspects of our design but are not utilized in subsequent iterations—these one-off prototypes leave behind a significant amount of waste. Unfortunately, currently, there are no established ways to re-purpose or minimize the waste generated in this process. Additionally, bulk order leftovers, under-optimized space management, and underdeveloped knowledge-sharing infrastructures are also contributing to sustainability issues in various aspects of rapid prototyping.

In this workshop (<http://www.sustainablemaking.com>), we aim to understand the sustainability challenges associated with rapid prototyping and seek to identify potential HCI research directions toward more sustainable "making" processes.

2 RELATED WORK

The HCI community is increasingly emphasizing the adoption of more environmentally-conscious approaches to digital fabrication [1, 2, 4, 10, 13], and more recently, even electronics re-use [11]. For instance, Sustainable Interaction Design (SID) [13] was proposed as a guideline to underscore the need for considering environmental factors and human behaviors when designing interaction technologies. Technical strategies have also been proposed for creating prototypes with a smaller ecological footprint, for example, via novel decomposable materials [5, 7, 15, 16], recycling from scrap artifacts [6, 11], or through easily deconstructable compositions [14, 18]. Moreover, emergent strategies have shown how modules and scrap parts can also be remodeled and re-manufactured, initiating a new lifecycle [8, 9, 11, 12, 17]. However, there is limited research on systematically applying sustainable practices within the framework of the rapid prototyping process, which is the focus of our workshop.

3 PLAN

The workshop will consist of keynotes, demos, hands-on building, and brainstorming sessions, divided into six distinct activities.

Activity 1 - Opening and short keynotes (9:00-10:00). Our workshop begins with a short introductory session, followed by three 20-minute keynotes. Potential topics range from research projects on new materials, tools, and systems for rapid prototyping, to their own perspectives on sustainable making. We expect to host three keynote speakers at the workshop.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3617433>

Activity 2 - Demo: Presenting current sustainability practice (10:00-11:00). Primed by the keynotes, the organizers will initiate a modified speed-dating activity [3]. In this activity, participants will meet in pairs or triads for 15 minutes before forming new groups. During these meetings, participants will present and demo their own hardware prototypes to each other, discussing the design process and any sustainability issues that arose during development. This format is designed to encourage maximum interaction among participants.

Activity 3 - Hands-on building: Reflecting on the prototyping process by making a phone stabilizer (11:00-12:30). In this activity, participants will be divided into groups of 4-5 people to conduct a hands-on mini-project: building a 2-axis phone camera gimbal. The organizers will provide each group with materials such as laser-cut acrylics, cardboard, Play-Doh, wire meshes, as well as basic crafting tools like scissors and hot glue. The goal is to encourage participants to reflect on this hands-on prototyping process and compare it with the more conventional digital prototyping processes they're used to. Participants will be asked to document and reflect on their prototyping process in a shared document. The organizers will facilitate a 15-minute reflection session. Each group will discuss their workflow and any potential sustainability issues encountered during the prototyping phase.

Activity 4 - Brainstorming: Sustainability challenges in making (13:30-15:00). Reflecting on discussions from Activities 2 and 3, a brainstorming session will focus on challenges and opportunities for sustainable rapid prototyping. The audience will be divided into small groups, with each group having a set of prototypes from Activity 3 and a hypothetical tangible innovation project to facilitate the brainstorming. The design process, potential prototyping tools, and material waste will be suggested as the key discussion points. In the final 30 minutes, each group will present the identified sustainability challenges, as well as the design opportunities for sustainable making tools, systems, or practices.

Activity 5: Brainstorming: Sustainability challenges beyond individual (15:00-16:00). Participants will be reshuffled into new groups for this brainstorming session. The key discussion points will be on project management, material management, and how physical materials, components, or prototypes can be reused on a larger scale. Participants are encouraged to reflect on their own lab or project management experience. Each group will present their findings in the final 15 minutes.

Activity 6: Summary and future plans (16:00-17:00). As the final activity, all participants will come together to map out the future of sustainable prototyping. Discussion items will include: 1) the list of key challenges to address, 2) the list of potential technical directions to pursue, 3) the methods to evaluate new sustainable practices, and 4) the approaches to promote sustainable prototyping on a large scale.

4 POST-WORKSHOP PLANS

The discussion from the workshop will be documented at <http://www.sustainablemaking.com>. Distilled results may also be shared with the HCI community via a position article. We also

plan to further build the community around sustainable making through the following mechanisms: 1) we will plan a new edition of this workshop either at CHI 2025 or at other venues such as the Dagstuhl seminar; 2) we will follow up with participants, seeking further online meetings and collaborations towards future research on this topic; 3) we may create monthly newsletters on recent research on this topic, sharing them with the participants and with the broader HCI and maker community.

5 ORGANIZERS

The organizing team represents the multidisciplinary nature of the workshop well.

Zeyu Yan is a Ph.D. student at the University of Maryland. He operates within the field of Personal Fabrication, Tangible User Interface, and Accessibility. Specifically, he is developing democratizable machines and processes that enable functional prototyping and tangible devices to assist blind users' access to modern technologies.

Tingyu Cheng, a senior Ph.D. student at Georgia Institute of Technology, has a background in Solid Mechanics and Human-Computer Interaction. His research involves the design, fabrication, and analysis of novel-material-based sensors and actuators, as well as their interactions with users.

Jasmine Lu is a Ph.D. student at the University of Chicago. Her work explores how the future of interactive technologies can be more sustainable. Her recent work explores making via reuse of e-waste. Additionally, she serves on the SIGCHI Sustainability Committee, focusing on sustainability in prototyping practices.

Pedro Lopes is an Associate Professor in Computer Science at the University of Chicago. Pedro focuses on integrating interfaces with the human body—exploring the interface paradigm that supersedes wearables. These include: muscle stimulation wearables that allow users to manipulate tools they have never seen before or that accelerate reaction time, or a device that leverages the smell to create an illusion of temperature.

Huaishu Peng is an assistant professor in the computer science department at the University of Maryland, College Park. He aims to advance interactive technologies by designing, prototyping, and evaluating novel artifacts that are personal and hands-on.

6 CALL FOR PARTICIPATION

We aim to understand the challenges of sustainable rapid prototyping and seek opportunities for improvement. We invite researchers, practitioners and makers interested in rapid prototyping and sustainable practices to join us for an in-depth discussion. Participants are greatly welcomed to bring hardware prototypes from previous and current projects. We also aspire to build a community and facilitate a sustained conversation as the concept of sustainability in rapid prototyping continues to evolve in the future.

REFERENCES

- [1] Eli Blevis. 2007. Sustainable interaction design: invention & disposal, renewal & reuse. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 503–512.
- [2] Hronn Brynjarsdottir, Maria Håkansson, James Pierce, Eric Baumer, Carl DiSalvo, and Phoebe Sengers. 2012. Sustainably unpersuaded: how persuasion narrows our vision of sustainability. In *Proceedings of the sigchi conference on human factors in computing systems*. 947–956.

- [3] Scott Davidoff, Min Kyung Lee, Anind K Dey, and John Zimmerman. 2007. Rapidly exploring application design through speed dating. In *UbiComp 2007: Ubiquitous Computing: 9th International Conference, UbiComp 2007, Innsbruck, Austria, September 16–19, 2007. Proceedings* 9. Springer, 429–446.
- [4] Carl DiSalvo, Phoebe Sengers, and Hrönn Brynjarsdóttir. 2010. Mapping the landscape of sustainable HCI. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 1975–1984.
- [5] Jeremy Faludi, Corrie M. Van Sice, Yuan Shi, Justin Bower, and Owen M.K. Brooks. 2019. Novel materials can radically improve whole-system environmental impacts of additive manufacturing. *Journal of Cleaner Production* 212 (2019), 1580–1590. <https://doi.org/10.1016/j.jclepro.2018.12.017>
- [6] Ollie Hanton, Zichao Shen, Mike Fraser, and Anne Roudaut. 2022. FabricatINK: Personal Fabrication of Bespoke Displays Using Electronic Ink from Upcycled E Readers. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 173, 15 pages. <https://doi.org/10.1145/3491102.3501844>
- [7] Elvin Karana, Davine Blauwhoff, Erik-Jan Hultink, and Serena Camere. 2018. When the Material Grows: A Case Study on Designing (with) Mycelium-based Materials. *International Journal of Design* 12, 2 (2018), 119 – 136. <https://search.ebscohost.com/login.aspx?direct=true&db=aft&AN=131581154&site=ehost-live>
- [8] Sunyoung Kim and Eric Paulos. 2011. Practices in the Creative Reuse of E-Waste. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 2395–2404. <https://doi.org/10.1145/1978942.1979292>
- [9] A. Kumar, K. Kumari, R. Sadasivam, and M. Goswami. 2021. Development of a 3D printer-scanner hybrid from e-waste. *International Journal of Environmental Science and Technology* 19, 3 (2021), 1447–1456. <https://doi.org/10.1007/s13762-021-03131-6>
- [10] Eldy S Lazaro Vazquez, Hao-Chuan Wang, and Katia Vega. 2020. Introducing the sustainable prototyping life cycle for digital fabrication to designers. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference*. 1301–1312.
- [11] Jasmine Lu, Beza Desta, K.D. Wu, Romain Nith, Joyce Passananti, and Pedro Lopes. 2023. ecoEDA: Recycling E-waste During Electronics Design. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA.
- [12] Ilan Mandel and Wendy Ju. 2023. Recapturing Product as Material Supply: Hoverboards as Garbatrage. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23). Association for Computing Machinery, New York, NY, USA, 564–579. <https://doi.org/10.1145/3563657.3596128>
- [13] Jennifer C Mankoff, Eli Blevis, Alan Borning, Batya Friedman, Susan R Fussell, Jay Hasbrouck, Allison Woodruff, and Phoebe Sengers. 2007. Environmental sustainability and interaction. In *CHI'07 extended abstracts on Human factors in computing systems*. 2121–2124.
- [14] Samar Sabie, Katherine W Song, Tapan Parikh, Steven Jackson, Eric Paulos, Kristina Lindstrom, Åsa Ståhl, Dina Sabie, Kristina Andersen, and Ron Wakkary. 2022. Unmaking@CHI: Concretizing the Material and Epistemological Practices of Unmaking in HCI. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 105, 6 pages. <https://doi.org/10.1145/3491101.3503721>
- [15] Katherine W Song, Aditi Maheshwari, Eric M Gallo, Andreea Danielescu, and Eric Paulos. 2022. Towards Decomposable Interactive Systems: Design of a Backyard-Degradable Wireless Heating Interface. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 100, 12 pages. <https://doi.org/10.1145/3491102.3502007>
- [16] Katherine Wei Song and Eric Paulos. 2023. Vim: Customizable, Decomposable Electrical Energy Storage. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 180, 18 pages. <https://doi.org/10.1145/3544548.3581110>
- [17] Ludwig Wilhelm Wall, Alec Jacobson, Daniel Vogel, and Oliver Schneider. 2021. Scrappy: Using scrap material as infill to make fabrication more sustainable. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (2021). <https://doi.org/10.1145/3411764.3445187>
- [18] Shanel Wu and Laura Devendorf. 2020. Unfabricate: Designing Smart Textiles for Disassembly. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376227>